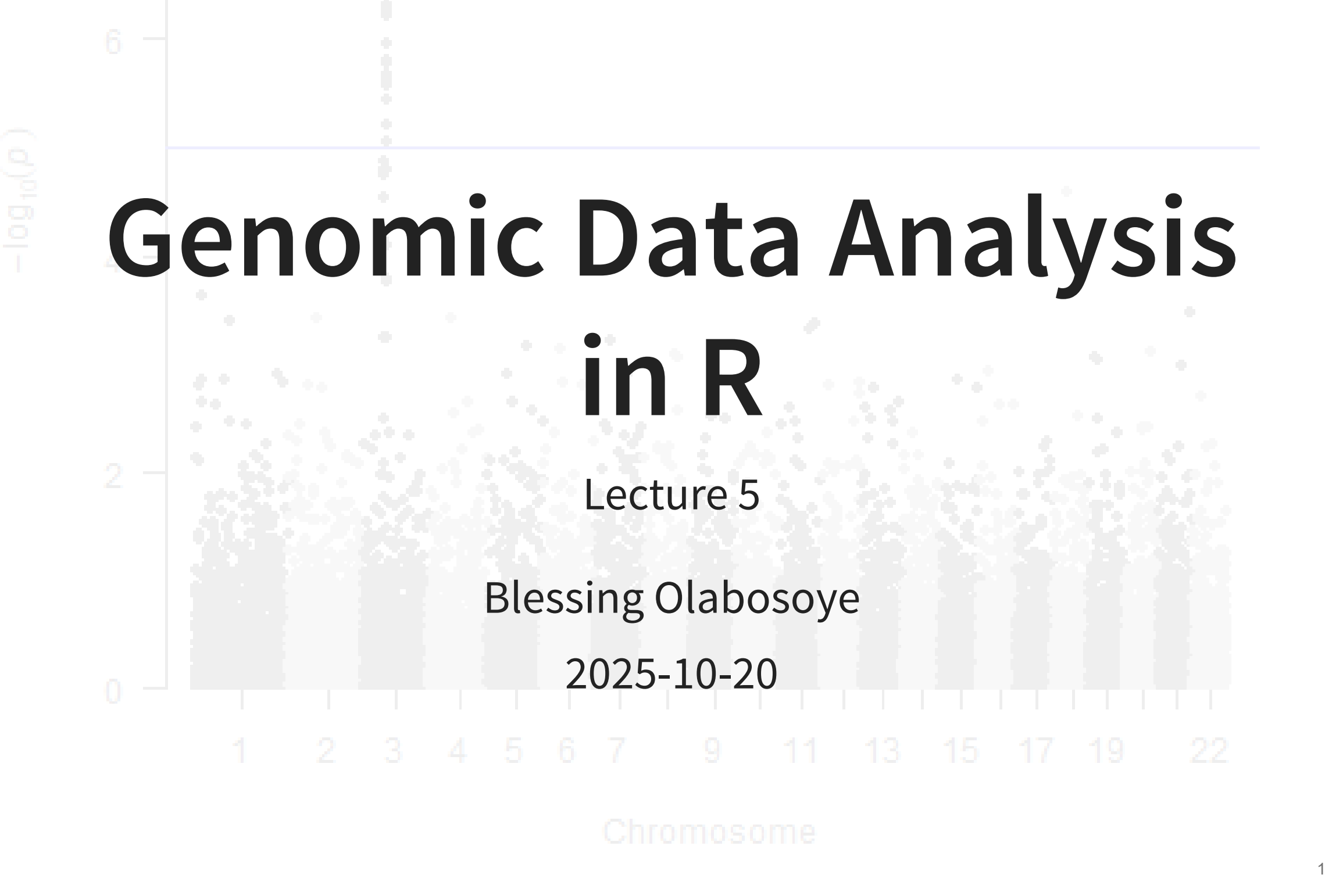




Genomic Data Analysis in R

Lecture 5

Blessing Olabosoye

2025-10-20

Learning outcomes

- Overview of polygenic score and genome-wide association study (GWAS)
- Understand and calculate polygenic score
- Understand and perform genome-wide association study (GWAS)
- Analyze and interpret GWAS plot
- Create a reproducible and publication-ready outputs

Linear Mixed Model

- Linear mixed model is a statistical model that accounts for both the fixed and random effects. It is linear because the predictor is a linear combination of all variable effects.

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{g} + \mathbf{e}$$

where, $\mathbf{y} \in \mathbb{R}^n$ is the phenotype of interest denoted as response variable

- Fixed effects
 - $\mathbf{X}$ is the $n \times p$ incidence/design matrix containing an intercept
 - $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of fixed effect(s) e.g. sex and batch on the phenotype

- Random effects
 - $\mathbf{Z}$ is the $n \times m$ incidence/design matrix linking
 - $\mathbf{g} \in \mathbb{R}^m$ the vector of random effect(s) e.g. allele substitution effect on the phenotype with a normal distribution $\mathbf{g} \sim \mathcal{N}(0, \sigma^2 \mathbf{g}\mathbf{I})$
 - $\mathbf{e} \in \mathbb{R}^n$ is the vector of residual variable with a normal distribution $\mathbf{e} \sim \mathcal{N}(0, \sigma^2 \mathbf{e}\mathbf{I})$

Note that β , $\mathbf{g}$, and $\mathbf{e}$ are unknown and will be estimated.

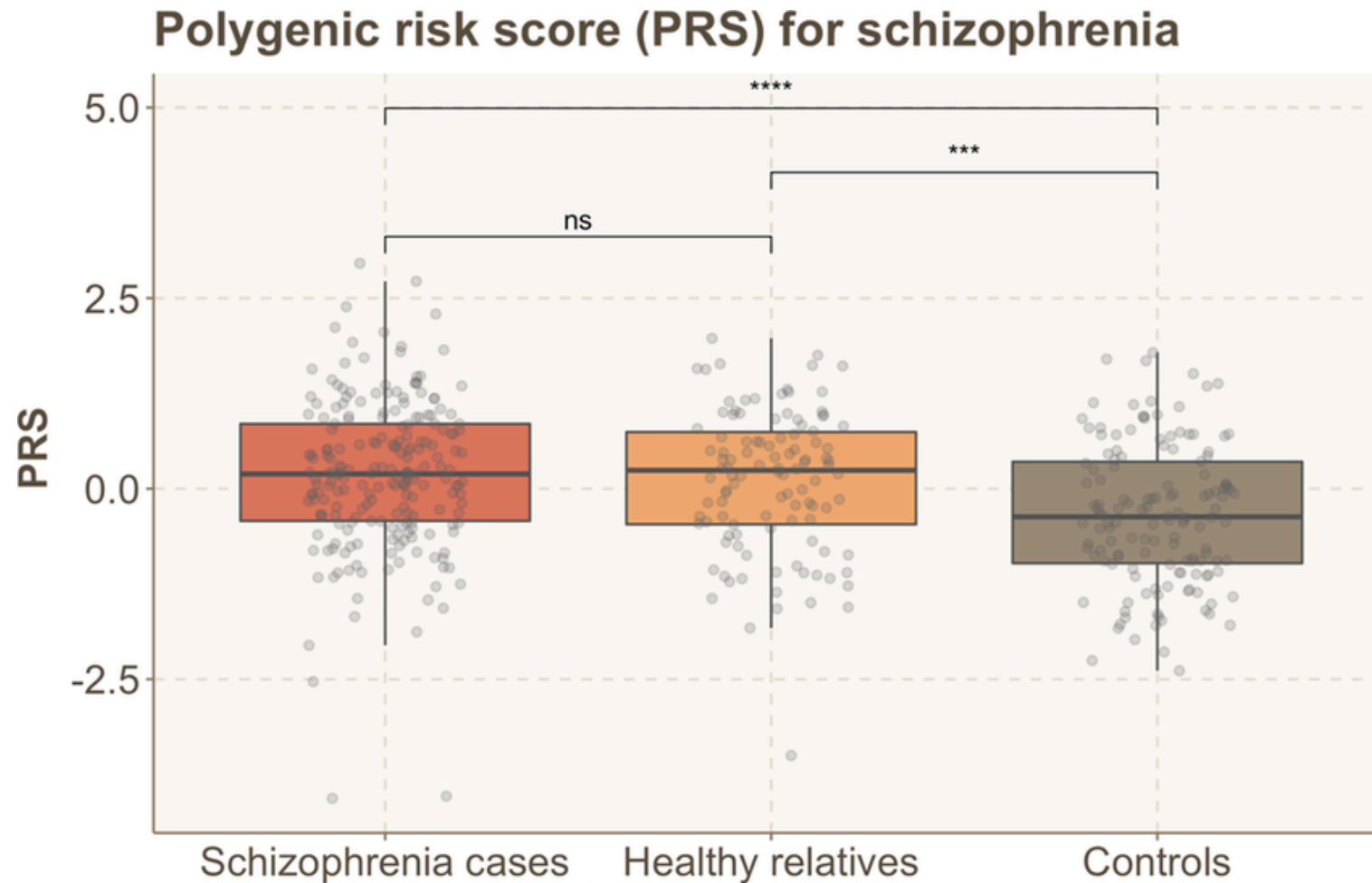
The estimate of β , $\hat{\beta}$, is called best linear unbiased estimator (BLUE) while the estimate of $\mathbf{g}$, $\hat{\mathbf{g}}$, is called best linear unbiased predictor (BLUP).

Polygenic score

- Polygenic score is a metric estimated using the genomic information of an individual and its association with a specific phenotype of interest. The “poly” indicates many and “genic” indicates gene (or genic) region; together it suggests a phenotype that is controlled and influenced by many genetic regions or loci across the genome.
- In human genetics, it is called polygenic risks score (PRS or PS) while in animal and plant genetics, it is genomic estimated breeding value (GEBV).
- It is computed using single-nucleotide polymorphism of individuals and their phenotype information. The statistic is used to obtain the score is:

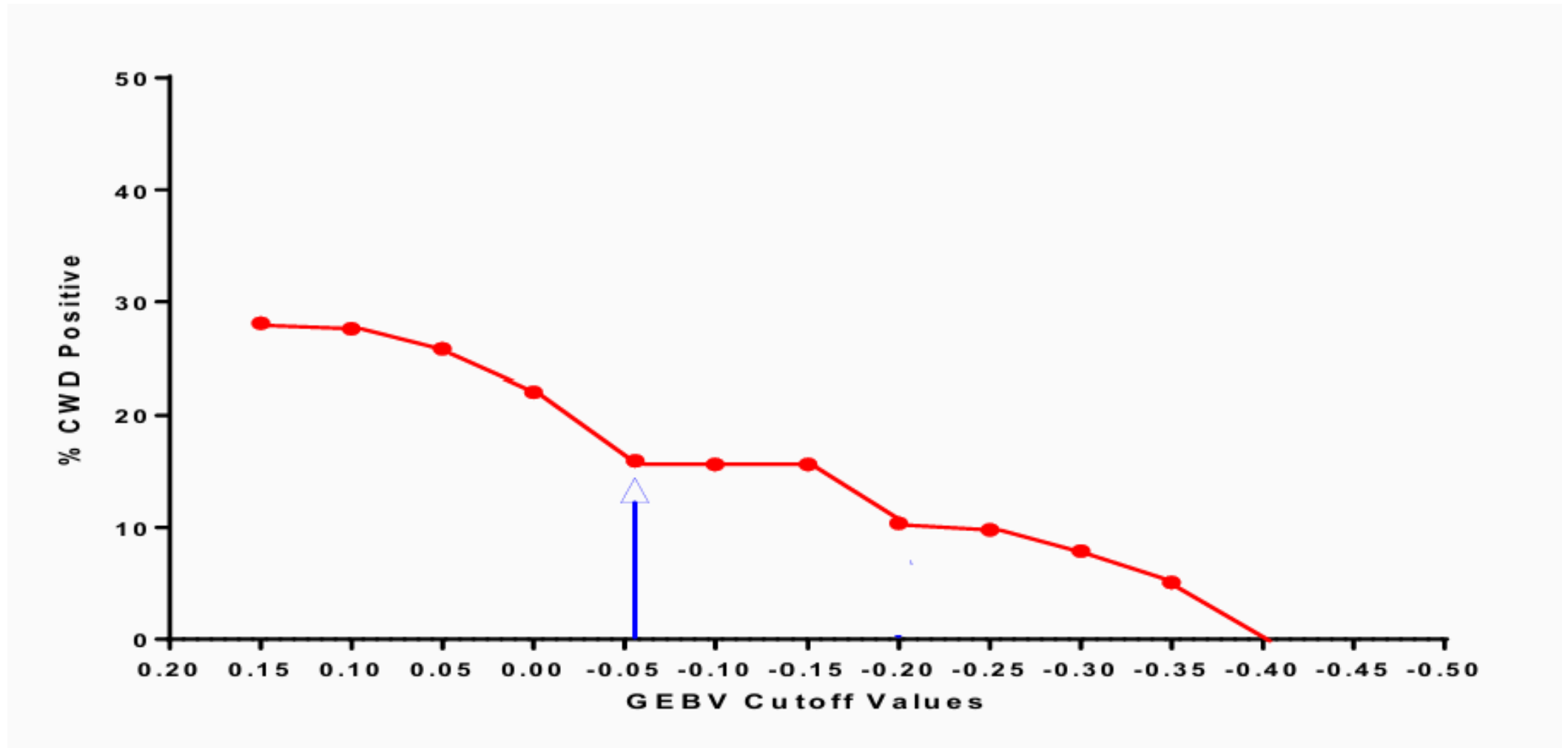
$$\hat{u} = \sum_{m=1}^M Z_i g_i$$

PRS in Schizophrenia research



Shaqiri et al. *Translational Psychiatry* (2022)12:529

GEBV in wild-tailed deer: chronic wasting disease (CWD)



WTD-CWD

Genome-wide association study

- Genome-wide association study (GWAS) is a study that uses high-throughput genomic information to associate or correlate genomic variants with a trait.
- While genomic (genotype) information is mostly used, nowadays transcriptomic (TWAS), proteomics (PWAS), and metabolomics (mGWAS) are also being used. Also, the genomic variants can be single-nucleotide polymorphisms (SNPs) or copy number variants.
- Although GWAS aims to identify associated variants and loci influencing a trait, it does not indicate **causality**. Post-GWAS steps, including fine-mapping and Mendelian randomization are needed to account for linkage disequilibrium and confounding effects.

GWAS: Model

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{S}\tau + \mathbf{W}\mathbf{u} + \mathbf{e}$$

where, $\mathbf{y} \in \mathbb{R}^n$ is the phenotype of interest denoted as response variable; $\mathbf{X}$ is the $n \times p$ incidence/design matrix linking $\beta \in \mathbb{R}^p$, the fixed effect(s) (e.g. sex) to the phenotype; $\mathbf{S}$ is the $n \times m$ incidence/design matrix linking $\tau \in \mathbb{R}^m$, the fixed additive SNP effect, to the phenotype; $\mathbf{W} = \mathbf{I}$ (if the genotyped individuals have phenotypic information) is the $n \times n$ incidence/design matrix linking $\mathbf{u} \in \mathbb{R}^n$, the random polygenic effect, to the phenotype; and $\mathbf{e}$ is the residual variable.

The $\mathbf{u}$ and $\mathbf{e}$ vectors follow a normal distribution: $\mathbf{u} \sim \mathcal{N}(0, \sigma^2 u \mathbf{K})$, $\mathbf{e} \sim \mathcal{N}(0, \sigma^2 e \mathbf{I})$.

GWAS: Hypothesis testing

$$H_0 : \tau_j = 0$$

$$H_a : \tau_j \neq 0$$

Test statistic for τ_j : $z = \frac{\hat{\tau}}{SE}$. If H_0 is rejected, then τ has a non-centrality parameter value, $NCP = (\hat{\tau} / SE)^2$

Hypothesis decision table

	H_0 is true	H_a is true
Fail to reject H_0	No association (α)	Type II error ($1 - \beta$)
Reject H_0	Type I error ($1 - \alpha$)	True association (β)

GWAS: Hypothesis testing

- In GWAS, we are performing statistical test that can range between 30,000 and 2,000,000 SNPs. Such large test results in multiple testing problem that increases the chance of making **Type I error**. To minimize the probability of making this error, several multiple testing correction can be employed to adjusting the significant threshold ($\alpha = 0.05$).
- However, there need to be a balance to optimally control both Type I and II error rate.

GWAS: Multiple testing corrections

- Bonferroni correction: Given a significant threshold ($\alpha = 0.05$), you can correct it by the number of independent test to perform. Therefore, the adjusted significant threshold becomes, $\alpha_{adj} = \frac{\alpha}{N}$, where N is the number of test e.g 2 million independent SNPs tests.

Example: Consider two populations for breast cancer with 1.5 or 2 million independent test

1. US: $0.05/1,500,000 = 3.3e^{-8}$; and 2) EU: $0.05/2,500,000 = 2e^{-8}$.

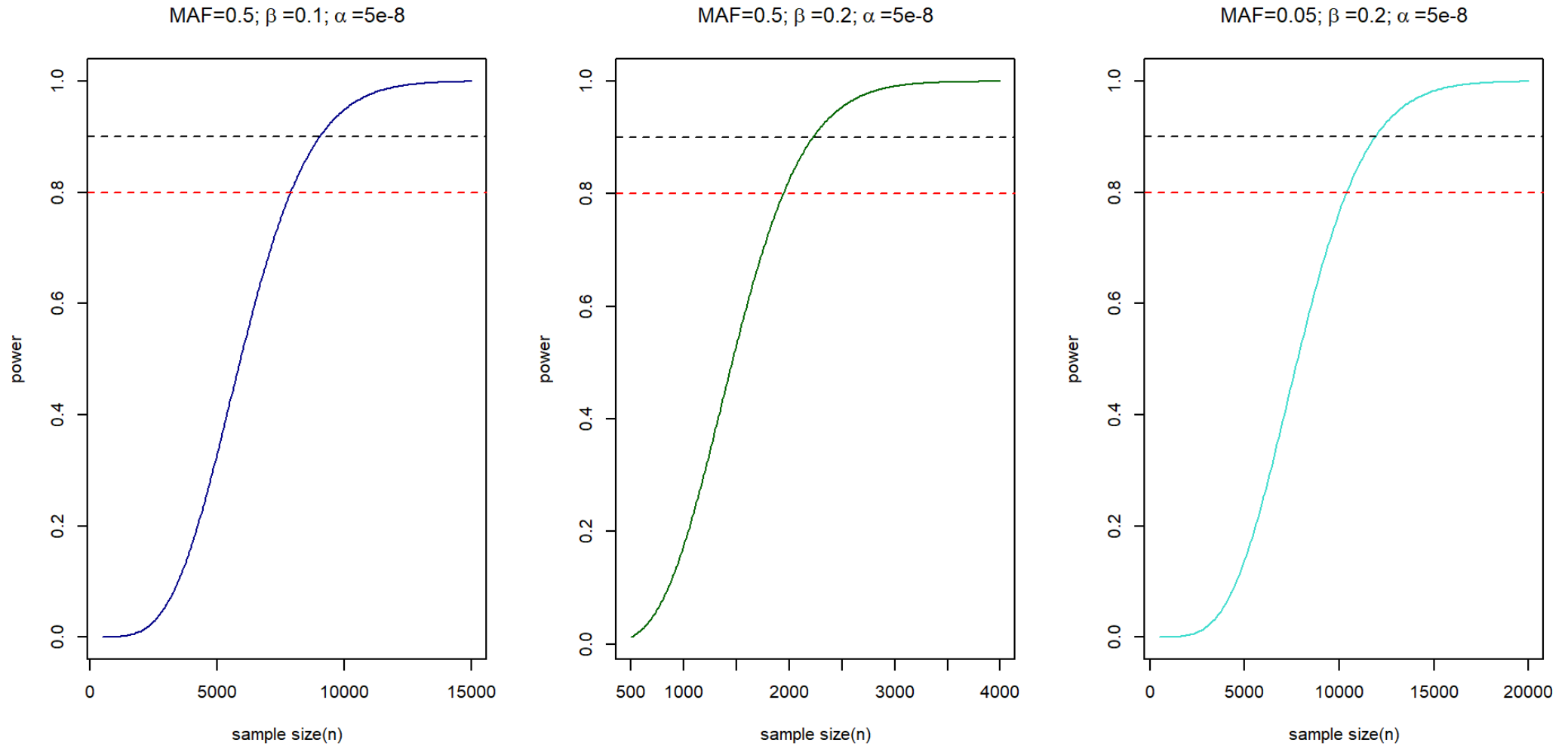
Due to this stringent correction, the risk of making Type II error increases.

- Benjamini-Hochberg FDR
- q-value (pFDR)

GWAS: Statistical power

- Linkage disequilibrium between SNPs and phenotype of interest
- Sample size (n)
- Effect size (β) or Proportion of phenotypic variance explained by the additive genetic effect (σ_g^2)
- Significant threshold (α)
- Minor allele frequency (MAF)

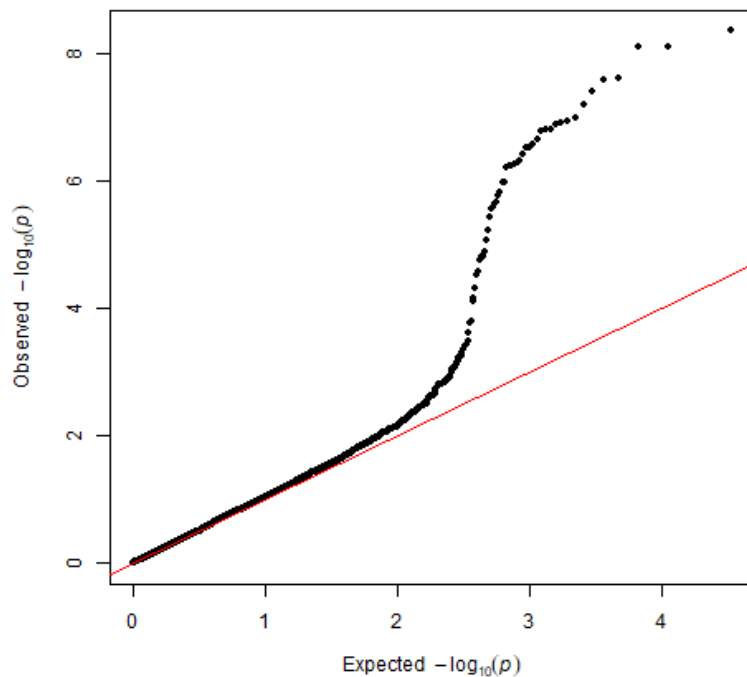
GWAS: Statistical power



Power calculation

GWAS Plotting

- Q-Q plot: This is a diagnostic plot to visualize how the observed p-values from GWAS relate to the null hypothesis p-values under uniform distribution. Generally, Q-Q plot is used to visualize if the quantile sample distribution comes from the quantile theoretical distribution.

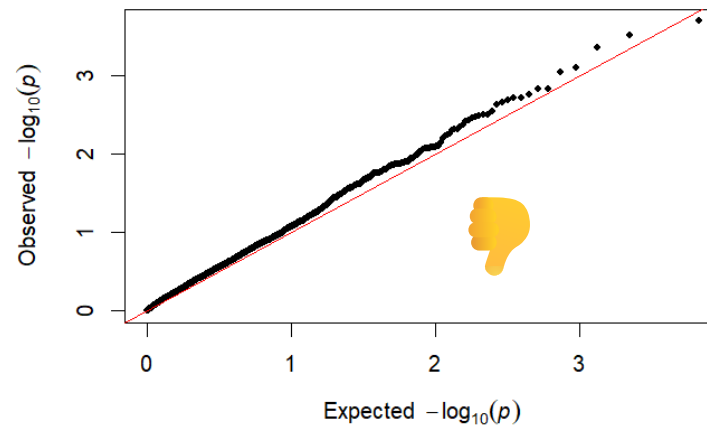
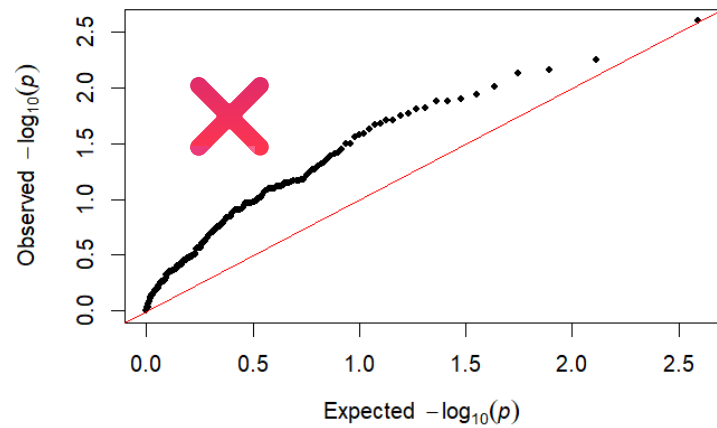
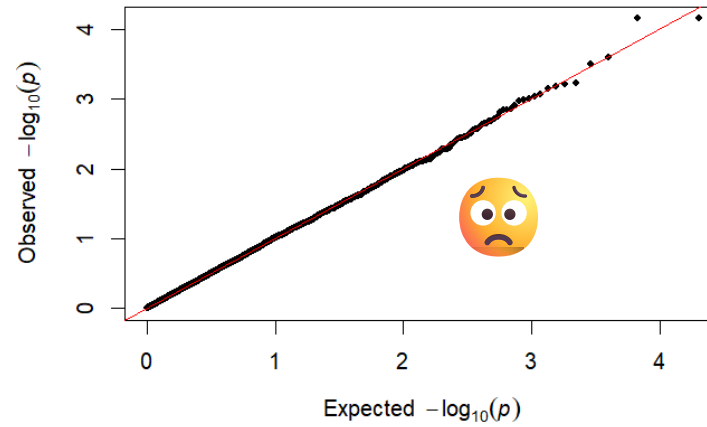
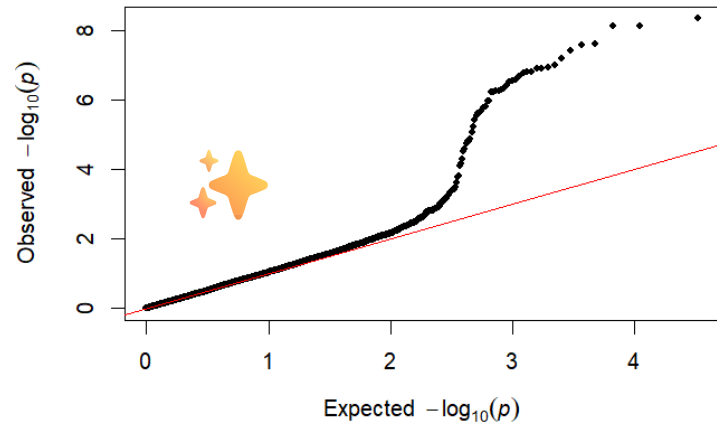


Any significant flare at the top-right corner from the main diagonal indicates that those points in the quantile sample distribution are not just by chance.

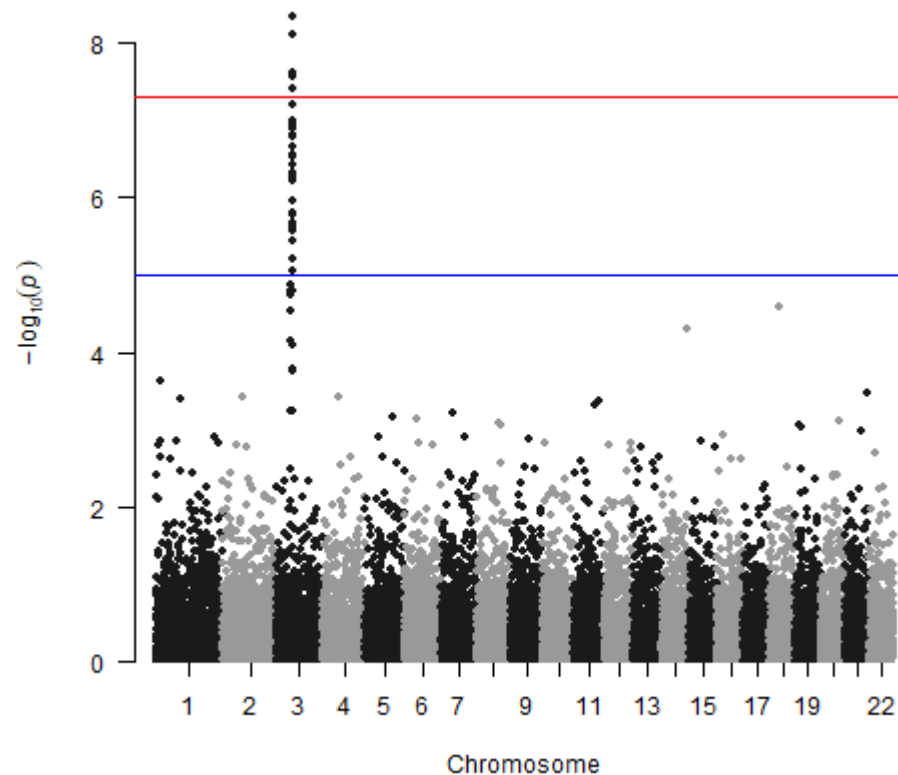
The genomic inflator factor (λ) is a good diagnostic to check the genome-wide inflation of the p-values. A deviation value away from 1 indicates systemic confounding in the data either inflation ($\lambda > 1$) or deflation ($\lambda < 1$)

Kinds of Q-Q plot to know

Q-Q plot of GWAS p-value

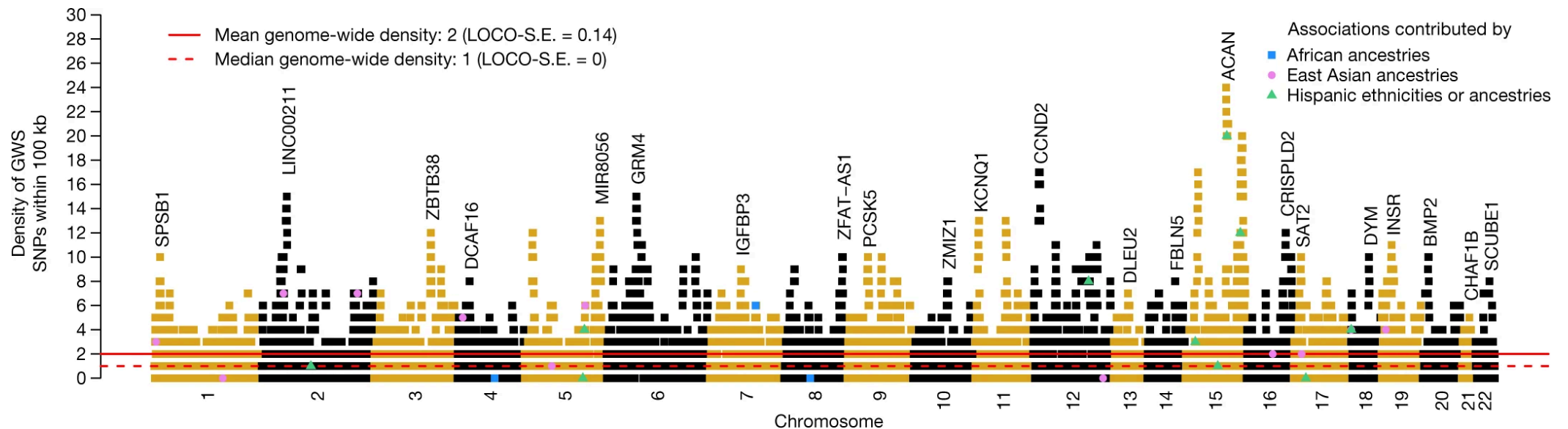


- Manhattan plot: This plot represents a scatter plot of the score of each variant per chromosome from a genome-wide association study. Manhattan plot can show the score as $-\log_{10} p$ value or proportion of additive genetic variance explained by SNPs.



- Brisbane plot: Like Manhattan plot represents a scatter plot of the score of each variant per chromosome from a genome-wide association study. Unlike Manhattan plot, Brisbane plot shows the score as the genomic density within a flanking region of a reference variant (e.g within 100kb).

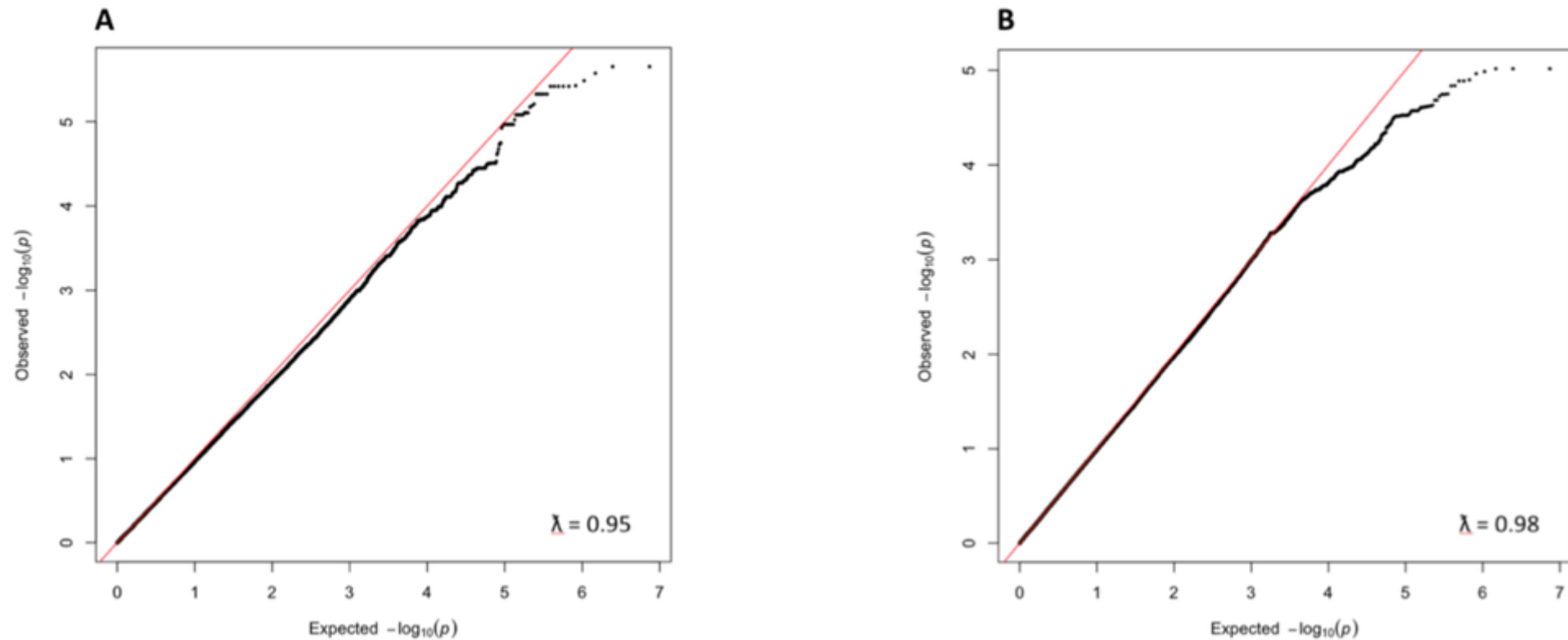
Human height



Yengo et al *Nature* (2022)610:704–712

Example: Schizophrenia - SOA25 and SOA5

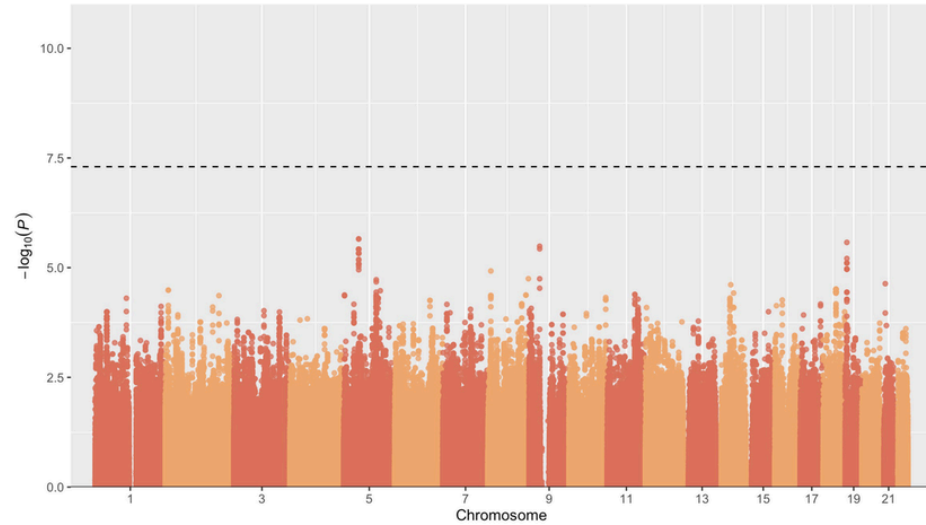
Q-Q plot



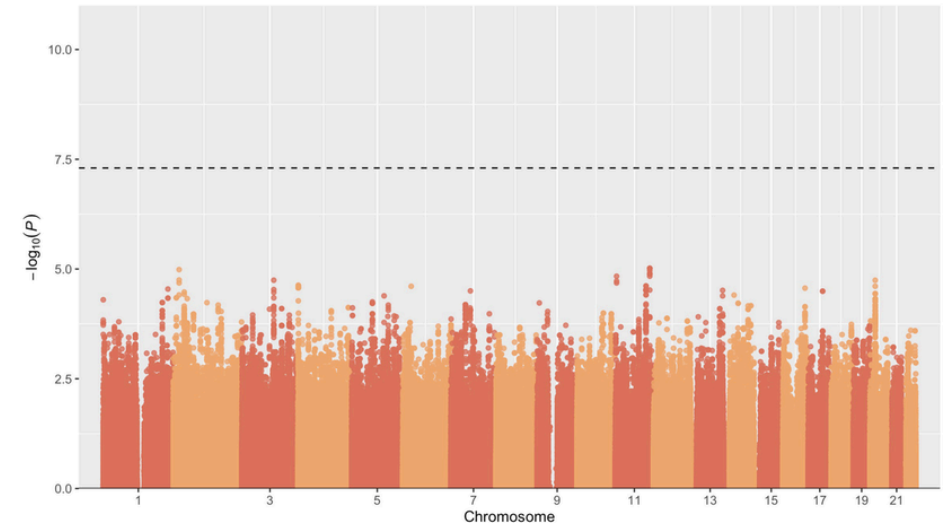
Shaqiri et al. *Translational Psychiatry* (2022)12:529

Manhattan plot: Schizophrenia - SOA25 and SOA5

A. SOA25



B. SOA5



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REVIEW

Open Access

The advantages and limitations of trait analysis with GWAS: a review

Arthur Korte^{*†} and Ashley Farlow[†]

Abstract

Over the last 10 years, high-density SNP arrays and DNA re-sequencing have illuminated the majority of the genotypic space for a number of organisms, including humans, maize, rice and *Arabidopsis*. For any researcher willing to define and score a phenotype across many individuals, Genome Wide Association Studies (GWAS) present a powerful tool to reconnect this trait back to its underlying genetics. In this review we discuss the biological and statistical considerations that underpin a successful analysis or otherwise. The relevance of biological factors including effect size, sample size, genetic heterogeneity, genomic confounding, linkage disequilibrium and spurious association, and statistical tools to account for these are presented. GWAS can offer a valuable first insight into trait architecture or candidate loci for subsequent validation.

Keywords: GWAS, *Arabidopsis*, Mixed model, Effect size, Genetic heterogeneity

Lab session!

